

Units and Dimensions in Physics

Detailed study notes covering physical quantities, SI units, fundamental and derived quantities, dimensions, dimensional formulae, dimensional analysis, unit conversion, limitations, examples, and exam revision.

1. Introduction

Units and dimensions are fundamental concepts used to measure and express physical quantities in physics.

$$\text{Physical Quantity} = \text{Numerical Value} \times \text{Unit}$$

Example: in 5 m, 5 is the numerical value and m is the unit.

2. Physical Quantities

A physical quantity is a measurable property of a physical system. Examples include length, mass, time, temperature, velocity, force, energy, and pressure.

Classification: Physical quantities are broadly divided into fundamental quantities and derived quantities.

3. Fundamental Quantities and SI Base Units

Quantity	SI Unit	Symbol	Dimension
Length	metre	m	[L]
Mass	kilogram	kg	[M]
Time	second	s	[T]
Electric current	ampere	A	[I]
Thermodynamic temperature	kelvin	K	[Θ]
Amount of substance	mole	mol	[N]
Luminous intensity	candela	cd	[J]

4. SI System of Units

The International System of Units (SI) is the internationally standardized system of measurement. Its seven base units are metre, kilogram, second, ampere, kelvin, mole, and candela.

5. Derived Quantities

Derived quantities are obtained by combining fundamental quantities mathematically.

$$\text{Area} = \text{length} \times \text{breadth} \rightarrow [L^2]$$

$$\text{Volume} = \text{length} \times \text{breadth} \times \text{height} \rightarrow [L^3]$$

$$\text{Velocity} = \text{distance/time} \rightarrow [LT^{-1}]$$

6. Dimensions

Dimensions describe how a physical quantity depends on fundamental quantities. Dimensions are written inside square brackets.

General dimensional form: $[M^a L^b T^c I^d \Theta^e N^f J^g]$

For mechanics, the most frequently used dimensions are M (mass), L (length), and T (time).

7. Unit vs Dimension

Unit	Dimension
Used to measure a physical quantity	Represents dependence on fundamental quantities
Examples: m, kg, s	Examples: [L], [M], [T]
Can change between unit systems	For a given physical quantity, dimensional formula is invariant
Has a numerical magnitude	Does not specify numerical magnitude

8. Dimensional Formulae of Common Quantities

Quantity	Definition/Formula	Dimensional Formula
Area	$A = l^2$	$[L^2]$
Volume	$V = l^3$	$[L^3]$
Velocity	$v = s/t$	$[LT^{-1}]$
Acceleration	$a = v/t$	$[LT^{-2}]$
Momentum	$p = mv$	$[MLT^{-1}]$
Force	$F = ma$	$[MLT^{-2}]$
Work	$W = Fd$	$[ML^2T^{-2}]$
Energy	E	$[ML^2T^{-2}]$
Power	$P = W/t$	$[ML^2T^{-3}]$
Pressure	$P = F/A$	$[ML^{-1}T^{-2}]$
Density	$\rho = m/V$	$[ML^{-3}]$
Frequency	$f = 1/T$	$[T^{-1}]$
Gravitational constant	$G = Fr^2/m^2$	$[M^{-1}L^3T^{-2}]$
Torque	$\tau = Fr$	$[ML^2T^{-2}]$
Angular velocity	$\omega = \theta/t$	$[T^{-1}]$
Angular acceleration	$\alpha = \omega/t$	$[T^{-2}]$
Moment of inertia	$I = mr^2$	$[ML^2]$
Surface tension	$S = F/l$	$[MT^{-2}]$
Dynamic viscosity	η	$[ML^{-1}T^{-1}]$
Young's modulus	$Y = \text{stress/strain}$	$[ML^{-1}T^{-2}]$

9. Important Derivations

9.1 Momentum

$$p = mv \rightarrow [p] = [MLT^{-1}]$$

9.2 Force

$$F = ma \rightarrow [F] = [MLT^{-2}]$$

9.3 Work and Energy

$$W = Fd \rightarrow [W] = [ML^2T^{-2}]$$

9.4 Power

$$P = W/t \rightarrow [P] = [ML^2T^{-3}]$$

9.5 Pressure

$$P = F/A \rightarrow [P] = [ML^{-1}T^{-2}]$$

9.6 Density

$$\rho = m/V \rightarrow [\rho] = [ML^{-3}]$$

9.7 Gravitational Constant

$$F = Gm_1m_2/r^2 \rightarrow [G] = [M^{-1}L^3T^{-2}]$$

10. Principle of Homogeneity of Dimensions

In a physically meaningful equation, every additive term must have the same dimensions.

$$s = ut + \frac{1}{2}at^2$$

Dimensions: $[s] = [L]$, $[ut] = [LT^{-1}][T] = [L]$, and $[at^2] = [LT^{-2}][T^2] = [L]$. Hence the equation is dimensionally homogeneous.

11. Dimensional Analysis

Dimensional analysis is a method of analyzing physical equations using dimensions of physical quantities.

Main applications: checking equations, deriving relationships, converting units, finding dimensions of unknown quantities, and detecting dimensional inconsistencies.

12. Deriving a Relation Using Dimensions

Suppose the time period T of a simple pendulum depends on length l and gravitational acceleration g .

$$T \propto l^a g^b$$

$$T = kl^a g^b$$

Using dimensions: $[T] = [L]^a [LT^{-2}]^b = L^{a+b} T^{-2b}$.

Comparing powers gives $a + b = 0$ and $-2b = 1$. Thus $b = -1/2$ and $a = 1/2$.

$$T = k\sqrt{l/g}$$

Dimensional analysis cannot determine the numerical constant k . For a simple pendulum under the usual small-angle approximation, the complete result is $T = 2\pi\sqrt{l/g}$.

13. Dimensionless Quantities

A dimensionless quantity has zero powers of all fundamental dimensions:

$$[M^0 L^0 T^0] = [1]$$

Examples include strain, relative density, coefficient of friction, refractive index, Mach number, angle, and solid angle. Plane angle is dimensionless in SI but is commonly expressed in radians.

14. Unit Conversion

When converting a quantity, the conversion factor must be raised to the appropriate power for area, volume, etc.

$$1 \text{ km} = 10^3 \text{ m}$$

$$1 \text{ cm} = 10^{-2} \text{ m}$$

$$1 \text{ mm} = 10^{-3} \text{ m}$$

$$1 \text{ cm}^2 = 10^{-4} \text{ m}^2$$

$$1 \text{ cm}^3 = 10^{-6} \text{ m}^3$$

$$1 \text{ m/s} = 3.6 \text{ km/h}$$

$$1 \text{ km/h} = 5/18 \text{ m/s}$$

15. Common SI Prefixes

Prefix	Symbol	Factor
kilo	k	10^3
mega	M	10^6
giga	G	10^9
tera	T	10^{12}
centi	c	10^{-2}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}

16. CGS, MKS and FPS Systems

CGS: centimetre, gram, second. **MKS:** metre, kilogram, second. **FPS:** foot, pound, second. The SI system is the internationally standardized system used today.

17. Important Derived SI Units

Quantity	SI Unit	Symbol
Force	newton	N
Energy/Work	joule	J
Power	watt	W

Quantity	SI Unit	Symbol
Pressure	pascal	Pa
Frequency	hertz	Hz
Electric charge	coulomb	C
Potential difference	volt	V
Resistance	ohm	Ω

18. Limitations of Dimensional Analysis

1. It cannot determine numerical constants such as 2, π , or $1/2$. 2. It cannot distinguish between physically different quantities having the same dimensions, such as work and torque. 3. It cannot determine functional forms such as sin, cos, exponential, or logarithmic dependence. 4. It cannot determine vector direction.

19. Exam-Focused Questions

1. Find the dimensional formula of force.
2. Find the dimensional formula of energy.
3. Find the dimensions of pressure.
4. Find the dimensions of gravitational constant G .
5. Find the dimensional formula of power.
6. Check whether $v^2 = u^2 + 2as$ is dimensionally correct.
7. Explain the principle of dimensional homogeneity.
8. Differentiate between units and dimensions.
9. Derive the dependence of a pendulum's time period on l and g using dimensional analysis.
10. State the limitations of dimensional analysis.

20. Quick Revision Sheet

Velocity: $[LT^{-1}]$

Acceleration: $[LT^{-2}]$

Momentum: $[MLT^{-1}]$

Force: $[MLT^{-2}]$

Work/Energy: $[ML^2T^{-2}]$

Power: $[ML^2T^{-3}]$

Pressure: $[ML^{-1}T^{-2}]$

Density: $[ML^{-3}]$

G : $[M^{-1}L^3T^{-2}]$

Angular velocity: $[T^{-1}]$

Moment of inertia: $[ML^2]$

Viscosity: $[ML^{-1}T^{-1}]$